

拡張カルマンフィルターを用いた 機械学習による地磁気永年変化推定

Estimation of Geomagnetic Secular Variation by Machine Learning with Extended Kalman Filter

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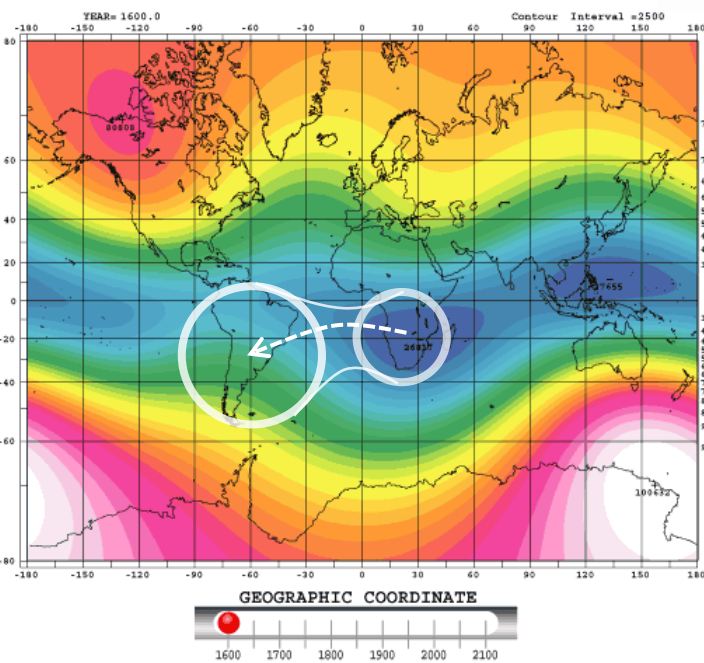
地球惑星科学専攻 地球物理学教室 太陽惑星系電磁気学分科

15:00 - 15:20, 2nd February (Mon) @ 修士論文発表会

INTRODUCTION | Background

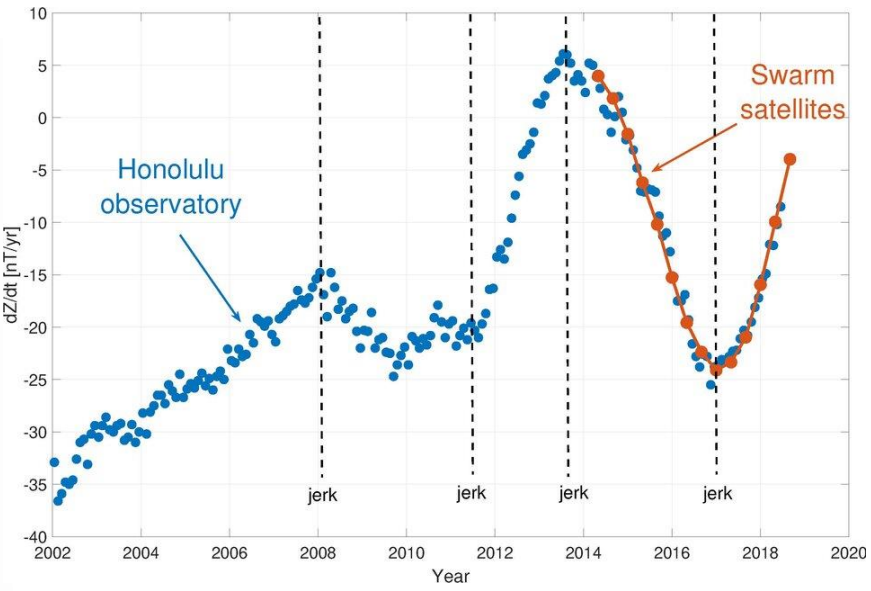
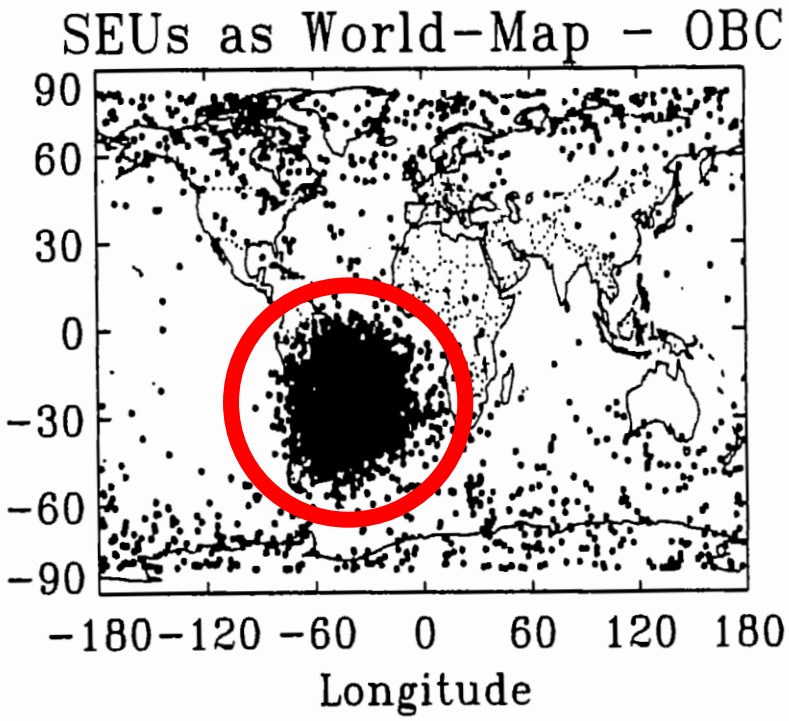
Geomagnetic *Secular Variation*

(Image: WDC for Geomagnetism, Kyoto*¹)



Location of UOSat-2 when *Single-event Upsets* were detected in its OBC memory during 1989

(Image: Daly et al, 1996*²)



Geomagnetic *Jerks* (= Sudden Acceleration)

(Image: ESA/DTU*³)

- **Forecasting the future evolution of the Earth's magnetic field (Secular Variation, SV)** is vital for applications ranging from space development to navigation.
- However, this prediction is a challenging problem due to **non-linear phenomena, such as geomagnetic jerks.**
- The **International Geomagnetic Reference Field (IGRF)** serves as a standardized field model updated every 5 years.

1. <https://wdc.kugi.kyoto-u.ac.jp>, 2. Daly et al (1996), "Space Environment Analysis: Experience and Trends" vol.392. p.15., 3. [Swarm helps explain Earth's magnetic jerks.](#)

INTRODUCTION | Variety of forecast methods

Two approaches for SV forecast



Math extrapolations.

E.g. = Linear / Polynomial extrapolation

Pros = Simple, data-driven

Cons = Hard to forecast non-linearity



Physics-based models.

E.g. = Geodynamo simulation /
Core surface flow modeling

Pros = Non-linear behavior

Cons = Computation cost

Physics-based SV candidates submitted to IGRF-13 (2020-25)

Institution	Method
BGS	Core surface flow modeling
IPGP	Geodynamo modeling
ISTerre	AR-2 process (COV-OBS.x2)
>> Japan <<	MHD dynamo simulation
Leeds Univ.	Core surface flow + Magnetic diffusion
Max Planck Inst.	3D-dynamo simulation
NASA/GSFC	Geodynamo model (NASA GEMS)
Univ. Potsdam	AR-2 process

OBJECTIVE : To develop a method for short-term (5-year) prediction of geomagnetic field change using **Machine Learning** (= **Data-driven** + **Non-linearity**).

METHOD | Recurrent Neural Network (RNN)

RNN architecture

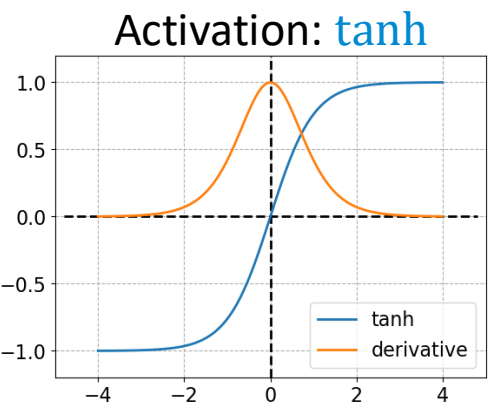
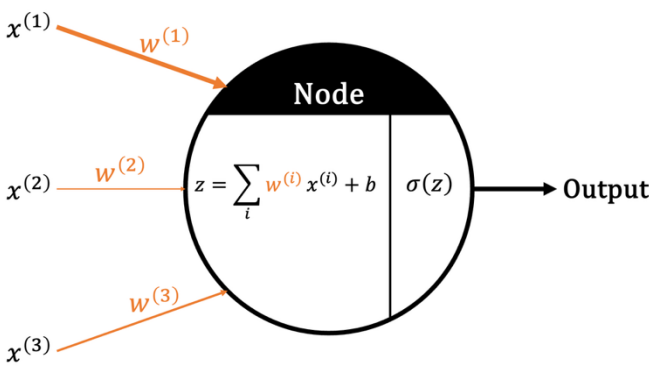
Hidden layer (recurrent)

$$\mathbf{h}(t_i) = \tanh(\mathbf{W}^{(\text{in})} \cdot \mathbf{x}(t_i) + \mathbf{W}^{(\text{rec})} \cdot \mathbf{h}(t_{i-1}) + \mathbf{b}^{(\text{rec})})$$

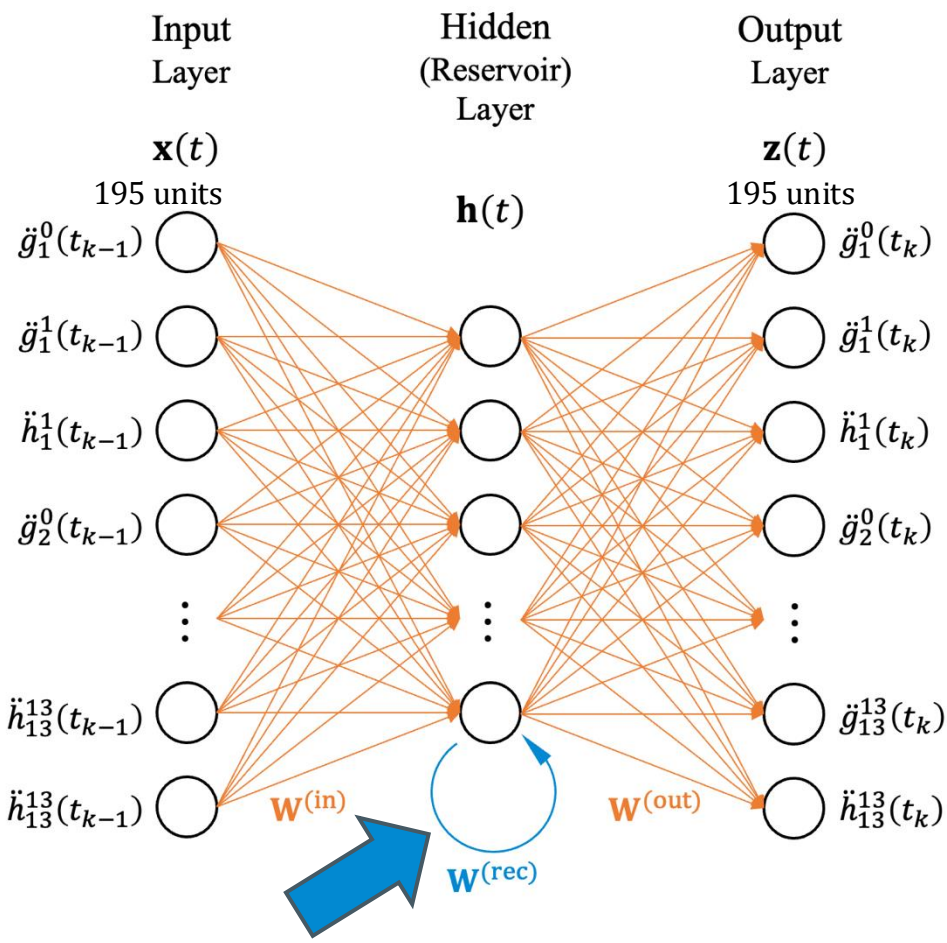
Output layer (affine)

$$\mathbf{z}(t_i) = \mathbf{W}^{(\text{out})} \cdot \mathbf{h}(t_i) + \mathbf{b}^{(\text{out})}$$

(input: $\mathbf{x}(t_i) = [\ddot{g}_n^m(t_{i-1})] \rightarrow$ output: $\mathbf{z}(t_i) = [\ddot{g}_n^m(t_i)]$).



Recurrent connection implicitly represents the structure in time



METHOD | EKF vs Backpropagation training scheme

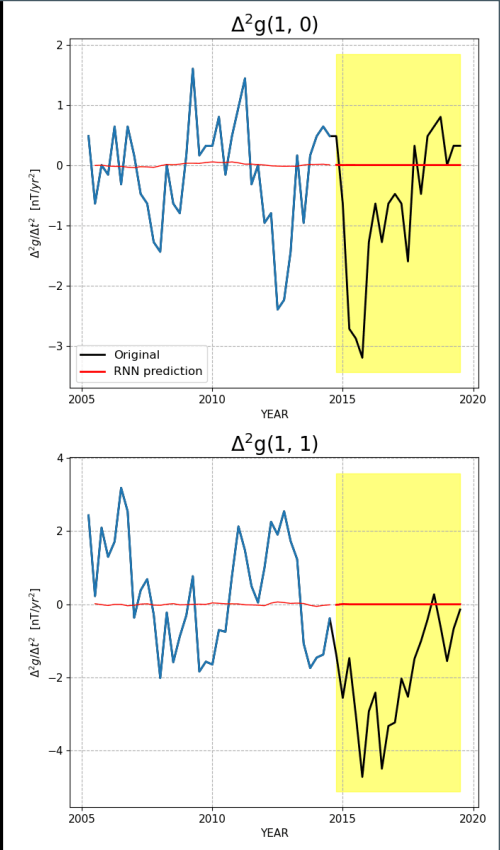
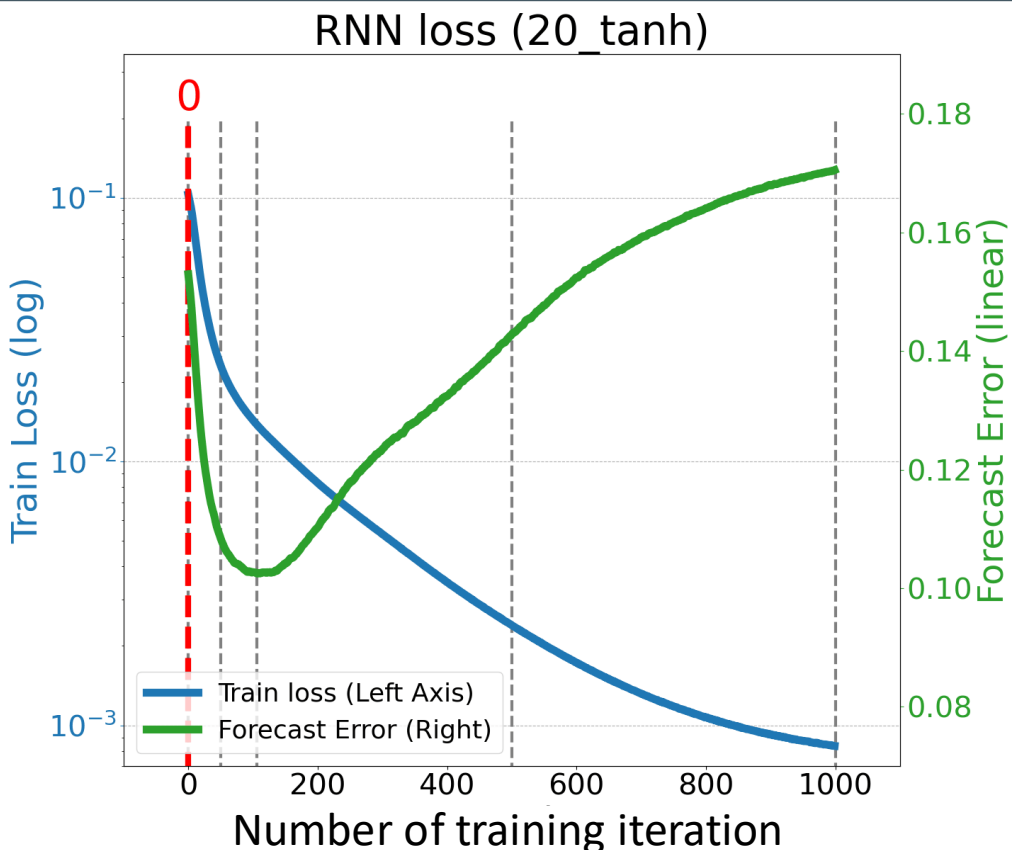
Backpropagation



Most widely used



Overfitting to the noise in training data



Extended Kalman Filter (EKF)



Observation error - **R**

→ Prevents **Overfitting**

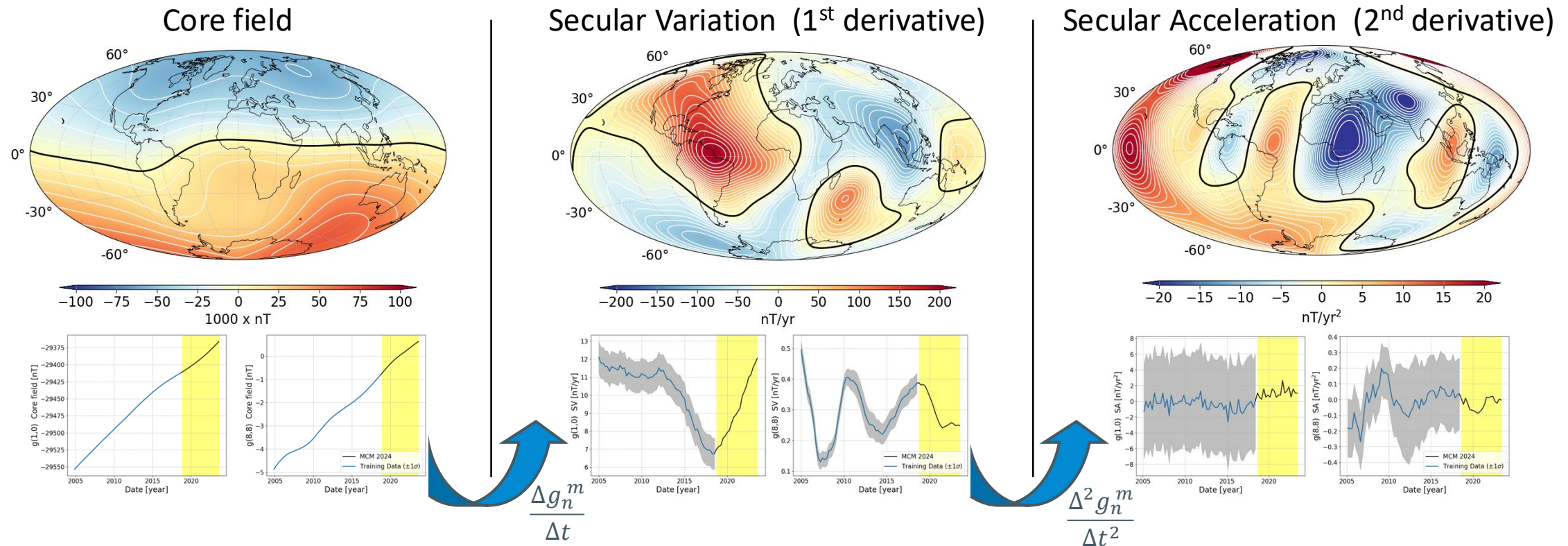


Computationally expensive

Backpropagation < EKF

<<< Dynamo Simulation

DATA | Geomagnetic field snapshots (MCM-2024)



Training data: Geomagnetic field snapshot provided by **MCM-2024 model (Ropp & Lesur 2023)**. Time series of **Gauss coefficients** up to **SH degree ≤ 13** every **3 months (0.25 years)**.

Derivatives: Incorporating the **time derivatives of the training data** into the learning process can lead to **enhanced accuracy**.

RESULTS | ML models vs Geodynamo simulation

Forecast error is measured

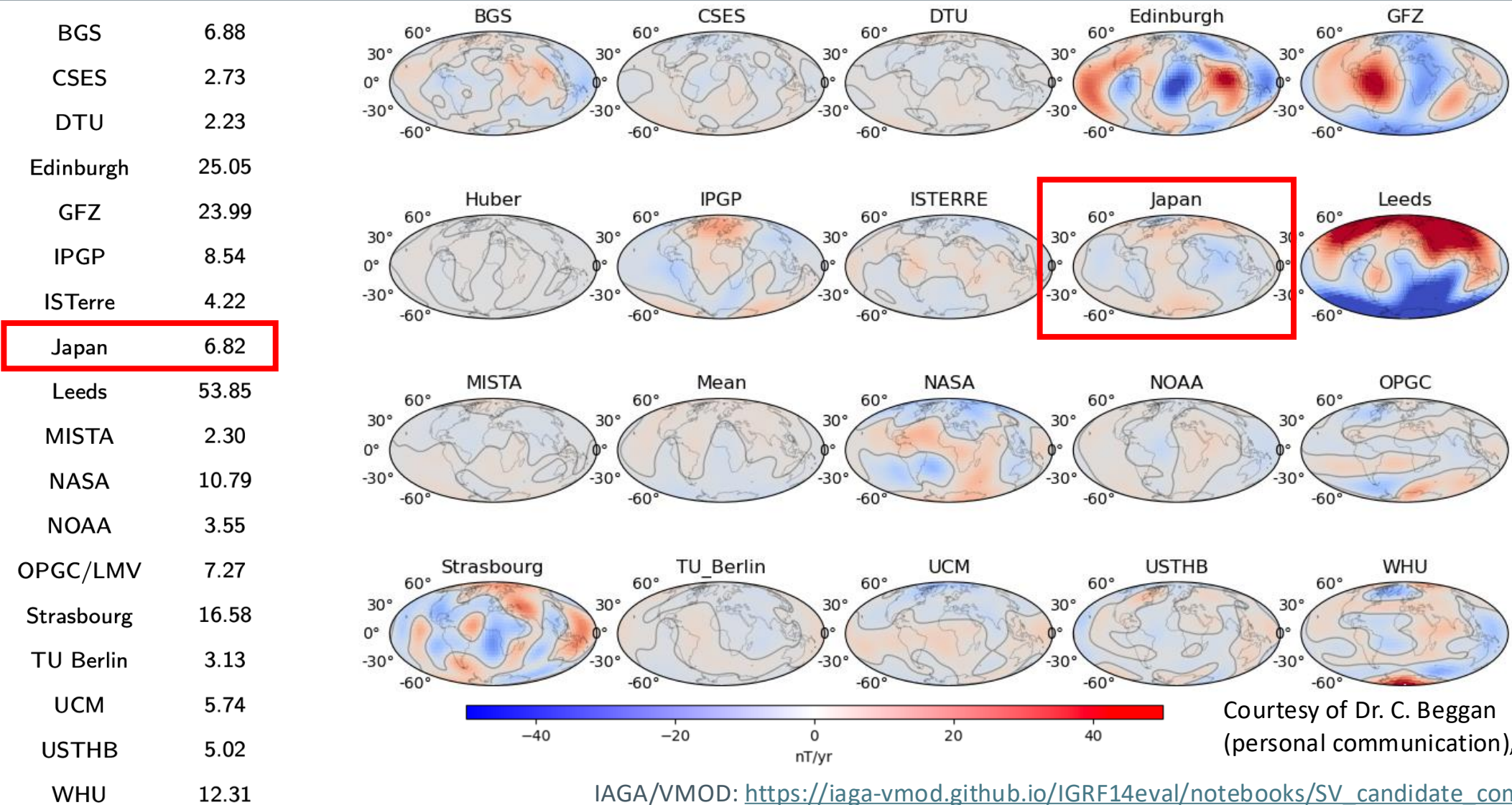
on the sqrt. of Energy:
$$dP(t) = \sum_{l=1}^{L_{MF}} \sum_{m=0}^l (l+1) \left[\left(g_l^{m(\text{model})}(t) - g_l^{m(\text{data})}(t) \right)^2 + \left(h_l^{m(\text{model})}(t) - h_l^{m(\text{data})}(t) \right)^2 \right].$$

Category	Method / Model	5-year Forecast Error [nT]
Classic ML Model	Auto Regression	88.1
	Vector Auto-Reg.	98.3
Neural Network Model	RNN (Backpropagation)	70.8 – 92.3
	RNN (Extended Kalman Filter)	70.3 – 84.1
	Echo State Network (Nakano et al., 2024)	80.0
Physics-based Simulation	Geodynamo x Data Assimilation (Minami et al., 2020)	100.9 – 228.5

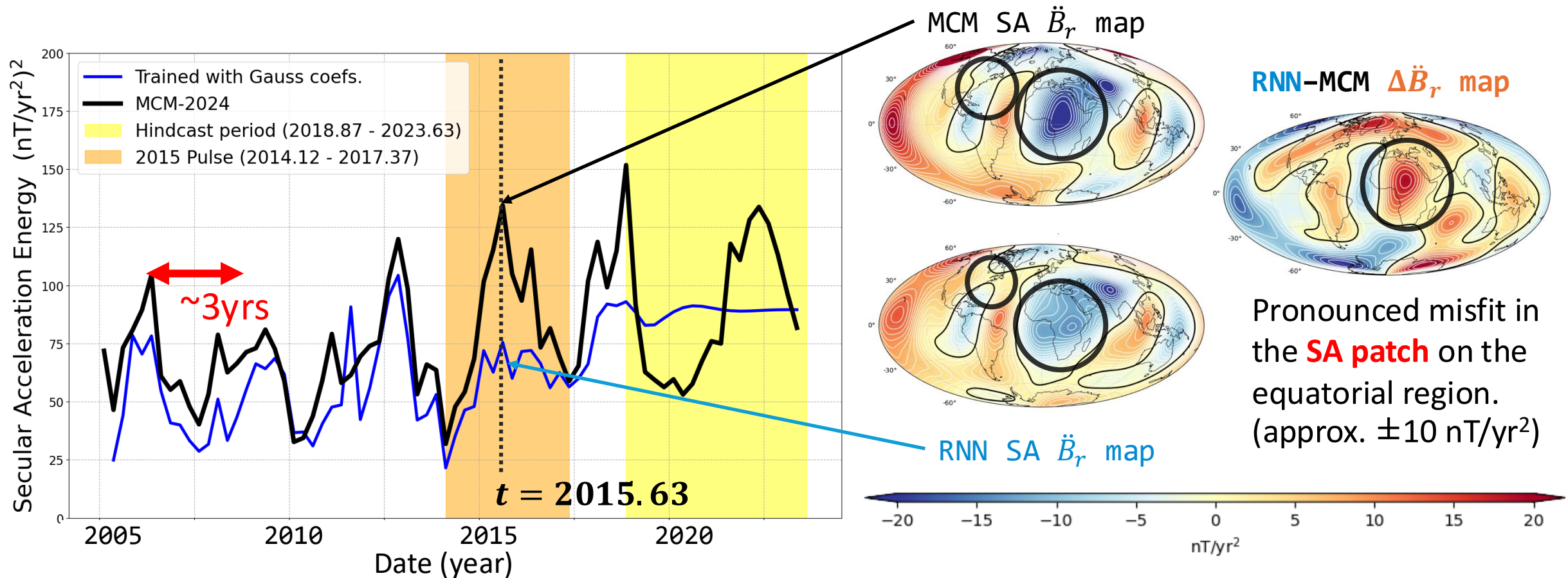
- EKF-trained RNN (EKF-RNN) achieved (1) the smallest best/worst error (ACCURATE) and (2) the smallest variance (ROBUST).

RESULTS | EKF-RNN Contribution to IGRF-14 SV (2025-30)

Difference in (left) sqrt. of energy [nT/yr] and (right) dBr/dt [nT/yr] of candidate SV models from the median model

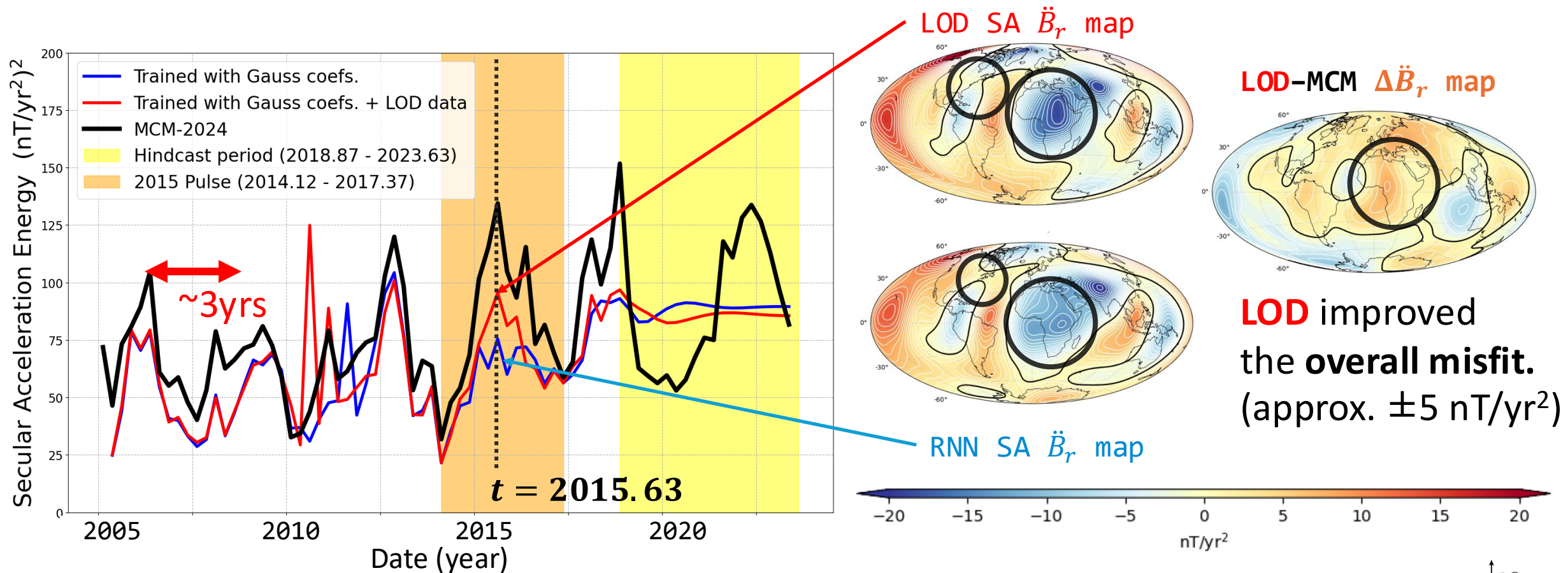


RESULTS | Training data = SA snapshots

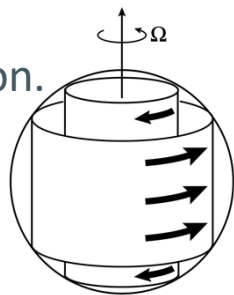


- **EKF-RNN** reproduces evolution of the **SA energy** with a clear **periodicity of approximately 3 years** (Lesur+ 2022).
- **EKF-RNN** failed to learn **SA pulse in 2015** (within training period) and SA patch on the equatorial region (Chulliat+ 2010; 2015).
- **The training misfit in 2015** may be attributed to the geomagnetic jerk that started early in 2014 (Torta+ 2015; Kotzé 2017).

RESULTS | Training data = SA snapshots + Length-Of-Day



- **EKF-RNN trained only with SA snapshots** failed to learn **SA pulse in 2015** and SA patch on the equatorial region.
- **LOD-augmented training (LOD-RNN)** improved the **training misfit at $t = 2015$ ($4.98 \rightarrow 2.43 \text{ nT/yr}^2$)**.
- This **Enhancement** may be attributed to the **common mechanism** of SA variation and LOD variation, such as **MHD waves inside the core**. (Dumberry & Bloxham 2004; Silva+ 2012; Holme & de Viron 2013)



CONCLUSIONS

➤ NEW METHOD

- ✓ We developed **EKF-RNN** as the third approach employing machine learning.
- ✓ **EKF-RNN** incorporates **error covariance matrices R** of training data and **prevents overfitting**.

➤ IGRF-14 CONTRIBUTION

- ✓ **EKF-RNN outperformed Physics-based approach** in 5-year hindcast experiment.
- ✓ **EKF-RNN's SV forecast for 2025-30** was submitted and contributed to **IGRF the 14th generation**.
- ✓ **EKF-RNN detected 3-yr quasi-periodicity** in the secular acceleration energy.

➤ LOD-AUGMENTED TRAINING

- ✓ The learning accuracy was further improved when the model was trained using **LOD data** in addition to the magnetic field data.
- ✓ **EKF-RNN** likely learned the correlation between **geomagnetic SA** and **LOD**, which accounts for the enhanced performance.
- ✓ **A machine-learning-based approach facilitates flexible incorporation of various data.**

CREDITS

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- Nakano, S., Sato, S. & Toh, H. Short-term prediction of geomagnetic secular variation with an echo state network. *Earth Planets Space* 76, 121 (2024). <https://doi.org/10.1186/s40623-024-02064-x>
- Sato, S., Nakano, S., Lesur, V. *et al.* Recurrent neural network trained with the extended Kalman filter to forecast the geomagnetic secular variation for IGRF-14. *Earth Planets Space* (2026). <https://doi.org/10.1186/s40623-025-02338-y>

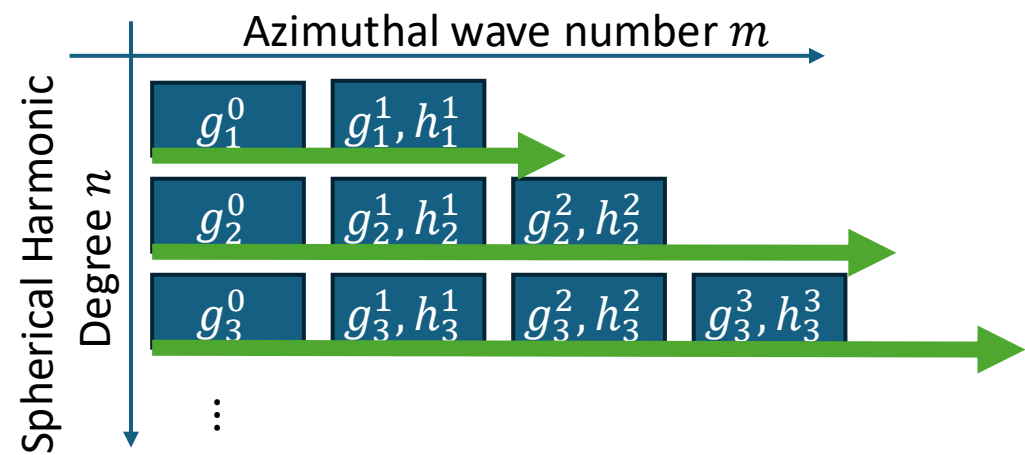
APPENDIX | IGRF-13 SV candidates in detail

Institution	Theme	Method
BGS	Physics	Core surface flow modeling (steady CSF with steady acceleration) fitted to Swarm + Ground Observations
UCM	Math	Swarm + Ground data at 2019.16 with Bootstrapping
NOAA/NCEI + CU	Math	Swarm data with Cubic splines knot at 2019.78
DTU Space	Math	Satellite [Ørsted, CHAMP, SAC-C, Cryosat-2, Swarm] + Ground data with 6 B-splines (CHAOS-7) knot at 2019.0
GFZ Potsdam	Math	Swarm + Ground data with 6 B-splines (Mag.num) knot at 2019.0
IPGP	Physics	Ensemble inverse geodynamo modelling with Swarm data and Kalman filter (MCM model)
ISTerre	Physics	Satellite [CHAMP, Swarm] + Ground data with Cubic B-splines governed by AR-2 process (COV-OBS.x2)
IZMIRAN	Math	Swarm data with parabolic approximation of gauss coefs. (2014.08 – 2019.08) extrapolated to 2022.5
>> Japan <<	Physics	MHD dynamo simulation assimilated to MCM model (by IPGP) with 4D-EnVar
Leeds Univ.	Physics	Core surface flow + Magnetic diffusion modeling fitted to CHAOS-6 model (by DTU Space)
Max Planck Inst.	Physics	3D-dynamo simulation (Parody; by IPGP) assimilated to COV-OBS.x1 (by ISTerre) and Kalmag (by Univ. Potsdam) with EnKF
NASA/GSFC	Physics	Geodynamo model (NASA GEMS) assimilated to gufm1 (by LeedsU), CM4 , and CM6 (by NASA) with EnKF
Univ. Potsdam	Physics	AR-2 process assimilated to Satellite [CHAMP, Swarm] data with Kalman filter smoothing (Kalmag model)
Univ. Strasbourg	Math	multi-variate singular spectrum analysis on Swarm + Ground data extrapolated to 2022.5 (C ³ FM2 model)

APPENDIX | Gauss coefs / Forecast Error Metrics

$$\begin{cases} \text{rot} \vec{B} = \vec{0} \\ \text{div} \vec{B} = 0 \end{cases} \iff \begin{cases} \vec{B} = -\text{grad } U \\ \text{div grad } U = -\nabla^2 U = 0 \end{cases}$$

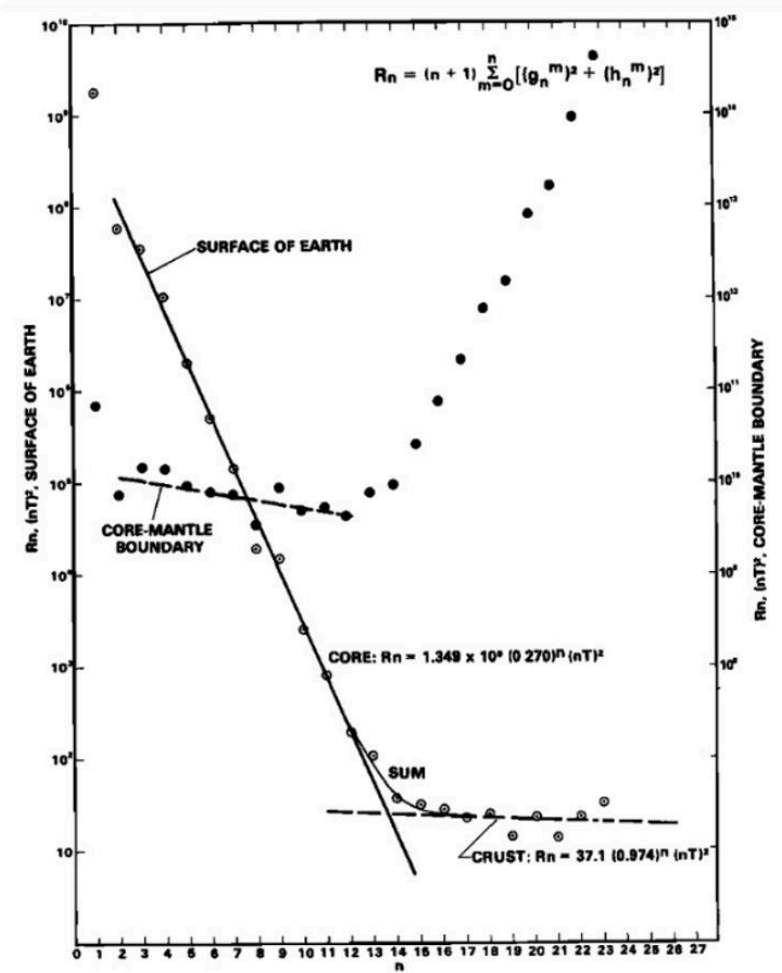
$$U(r, \theta, \phi) = R_E \sum_{n=1}^{\infty} \sum_{m=0}^n \left\{ (g_n^m \cos m\phi + h_n^m \sin m\phi) \left(\frac{R_E}{r} \right)^{n+1} \right\} P_n^m(\cos \theta) \\ + R_E \sum_{n=1}^{\infty} \sum_{m=0}^n \left\{ (q_n^m \cos m\phi + s_n^m \sin m\phi) \left(\frac{r}{R_E} \right)^n \right\} P_n^m(\cos \theta)$$



Degree dependence of Error : $\sqrt{dP_n} = \sqrt{\sum_m (n+1) [g_{\text{TEST}}^m - g_{\text{PRED}}^m]^2}$

Total Error : $\sqrt{dP} = \sqrt{\sum_n dP_n}$

Mauersberger-Lowes spectrum
(Langel and Estes, 1982)



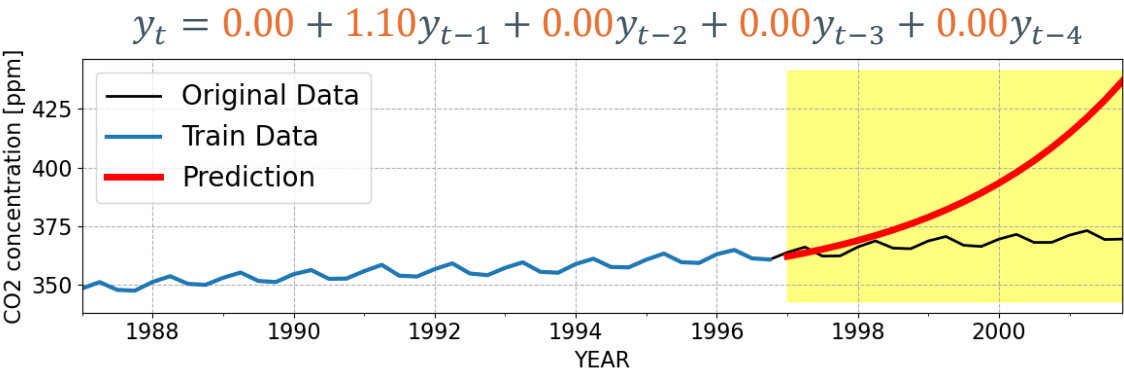
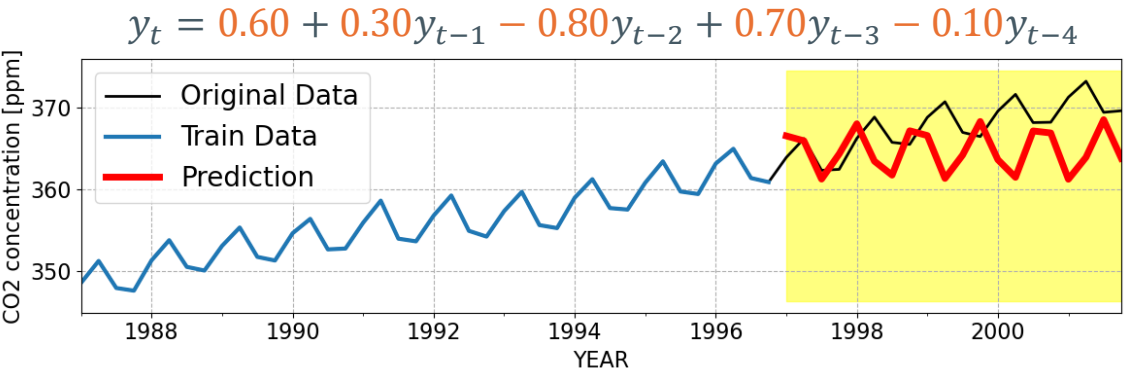
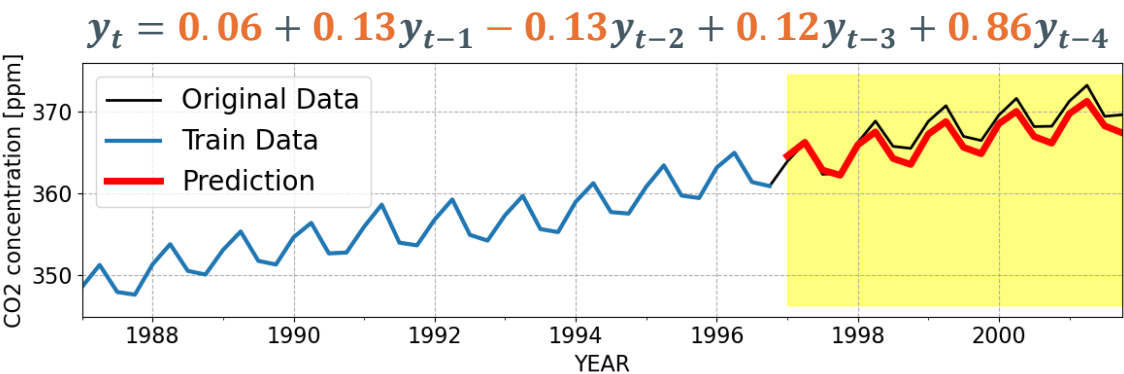
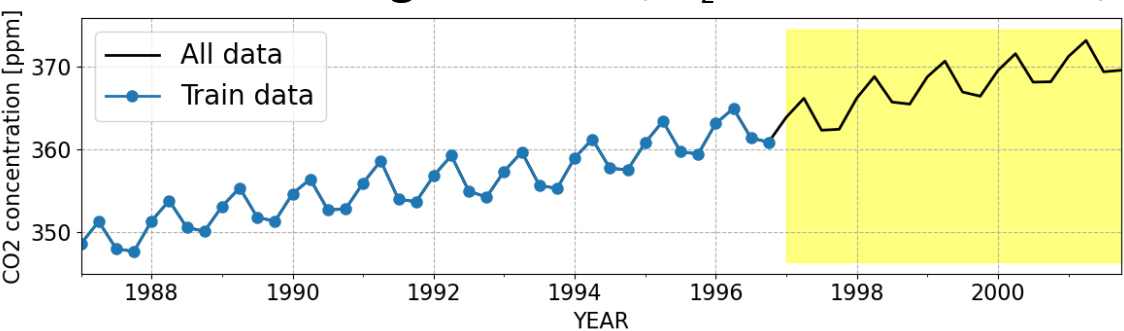
APPENDIX | What is ML training? = Parameter estimation

Example: AR(4) model

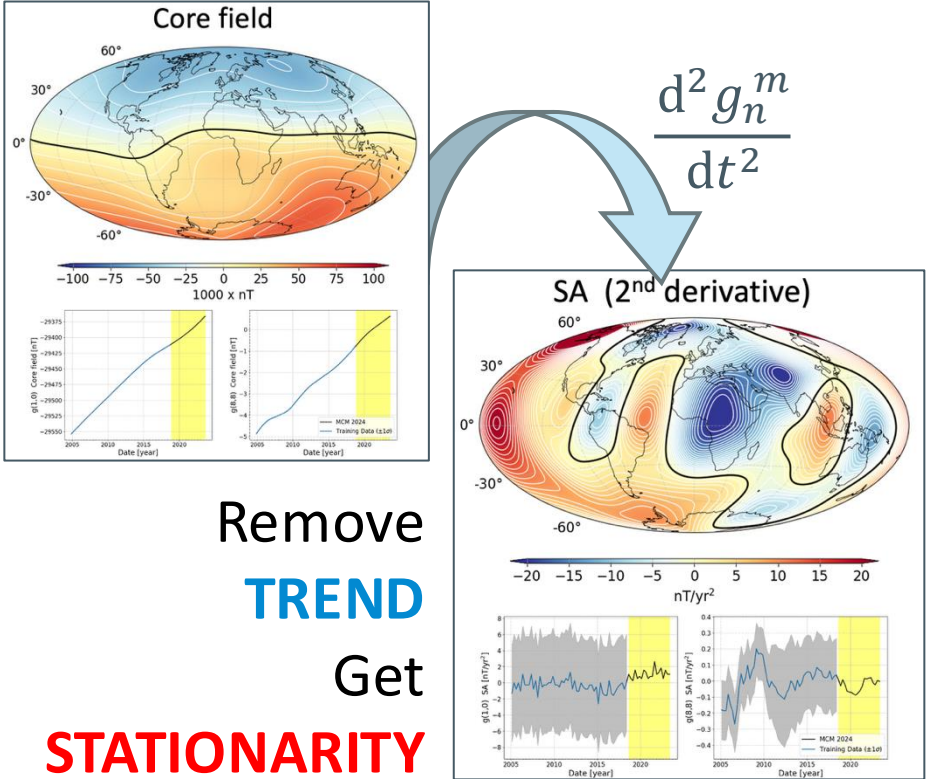
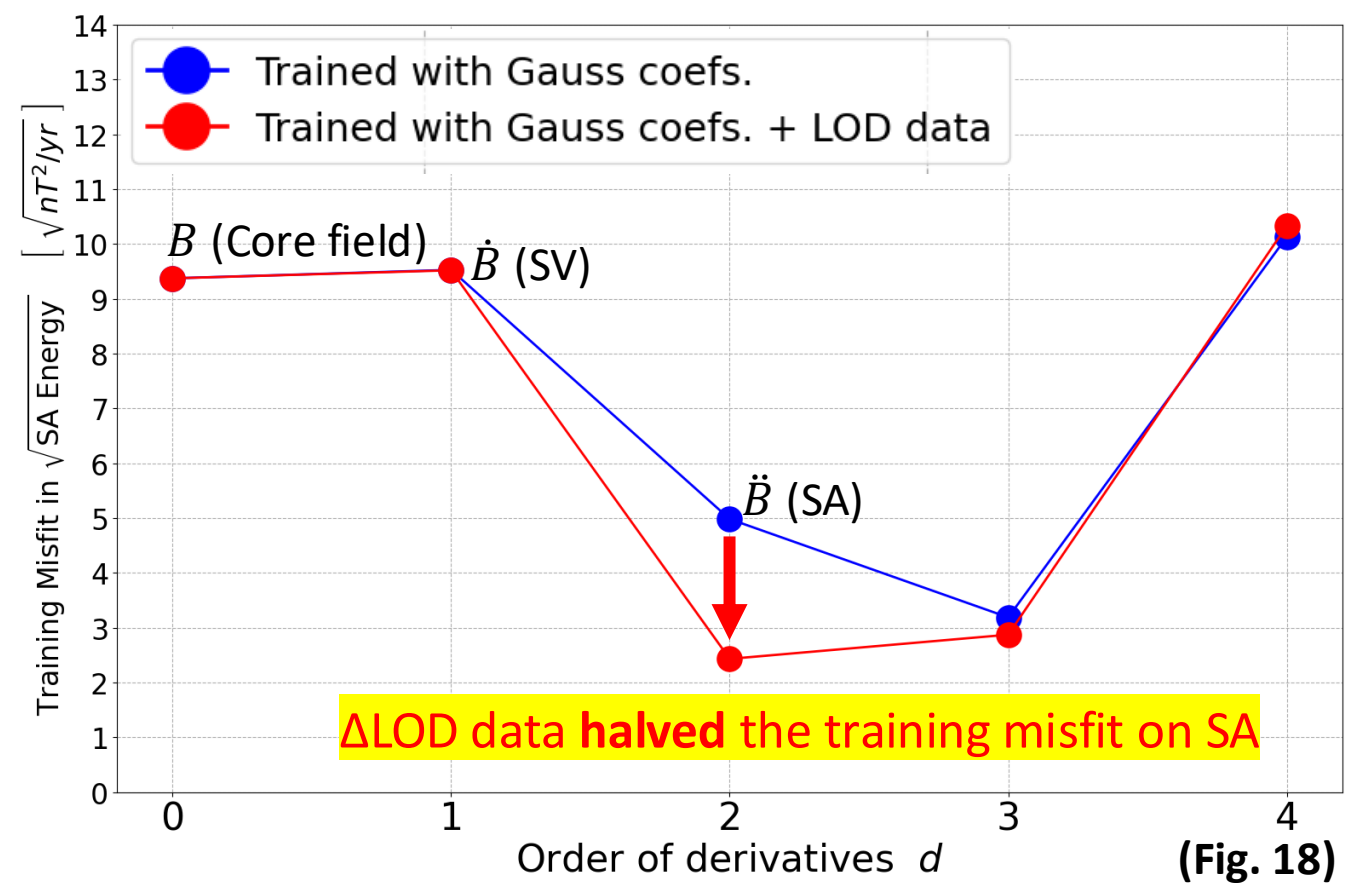
$$y_t = b + w_1 y_{t-1} + w_2 y_{t-2} + w_3 y_{t-3} + w_4 y_{t-4}$$

(Trainable Parameter: b (bias), $w_1 \dots w_4$ (weight))

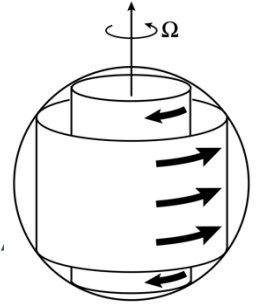
Original Data (CO₂ at Mauna Loa, Hawaii)



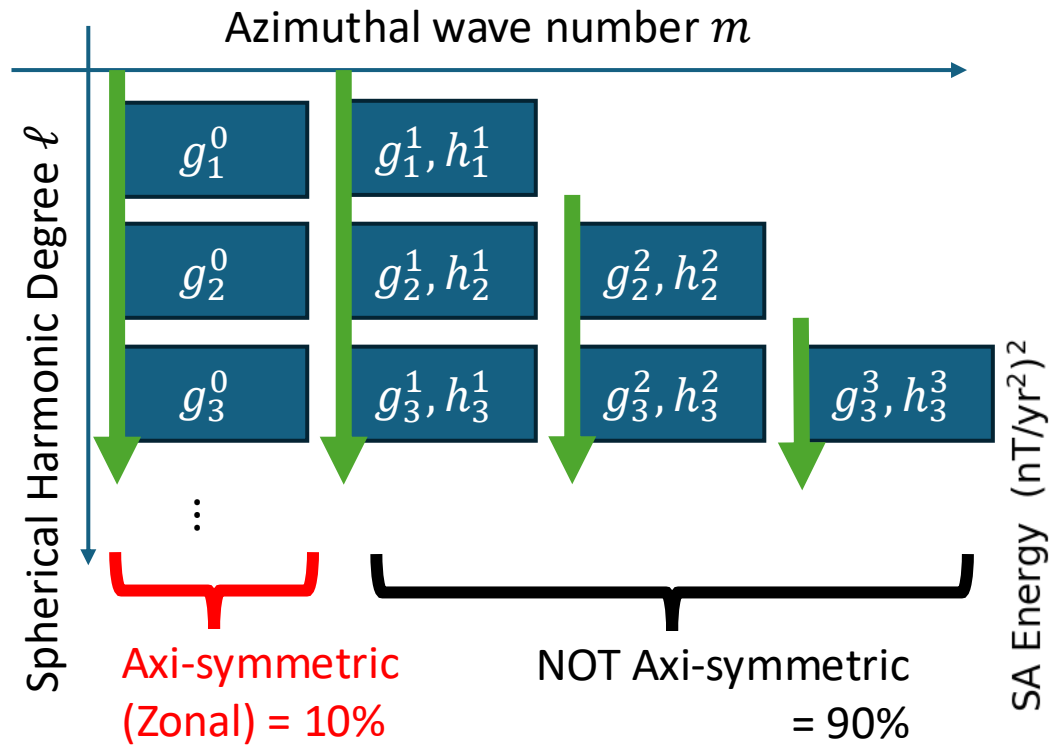
RESULTS | LOD's selective enhancement on SA



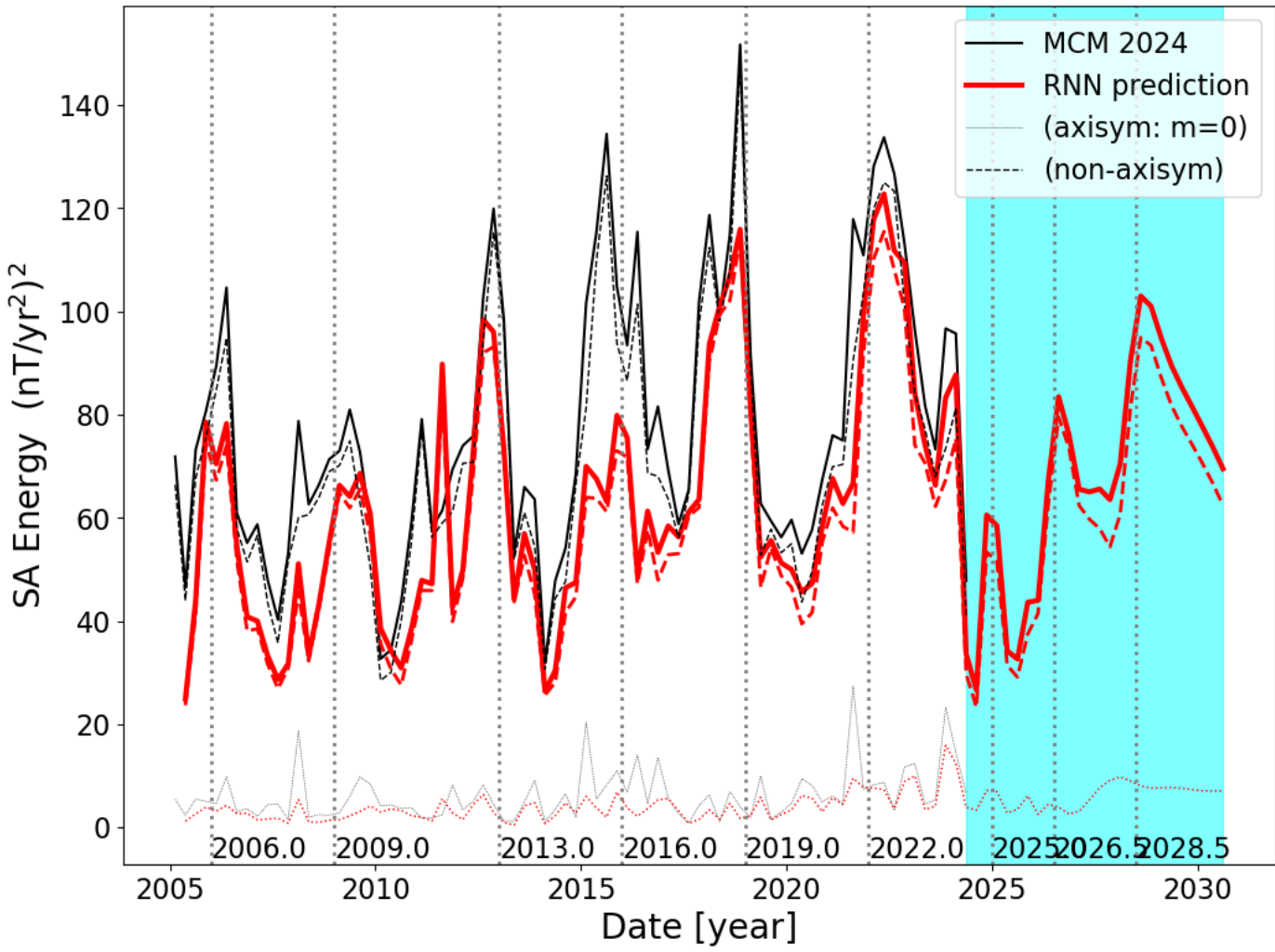
- The training misfit decreases with the **2nd (SA)** and **3rd derivatives** of training data.
- **ΔLOD data** may have aided EKF-RNN in the learning and prediction of **only the SA dynamics**.
- This **Selective Enhancement** may be attributed to the **common mechanism** of SA variation and LOD variation, such as **MHD waves inside the core**. (Dumberry & Bloxham 2004; Silva+ 2012; Holme & de Viron 2013)



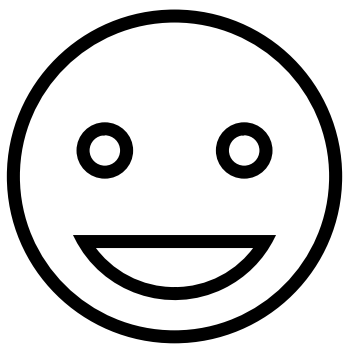
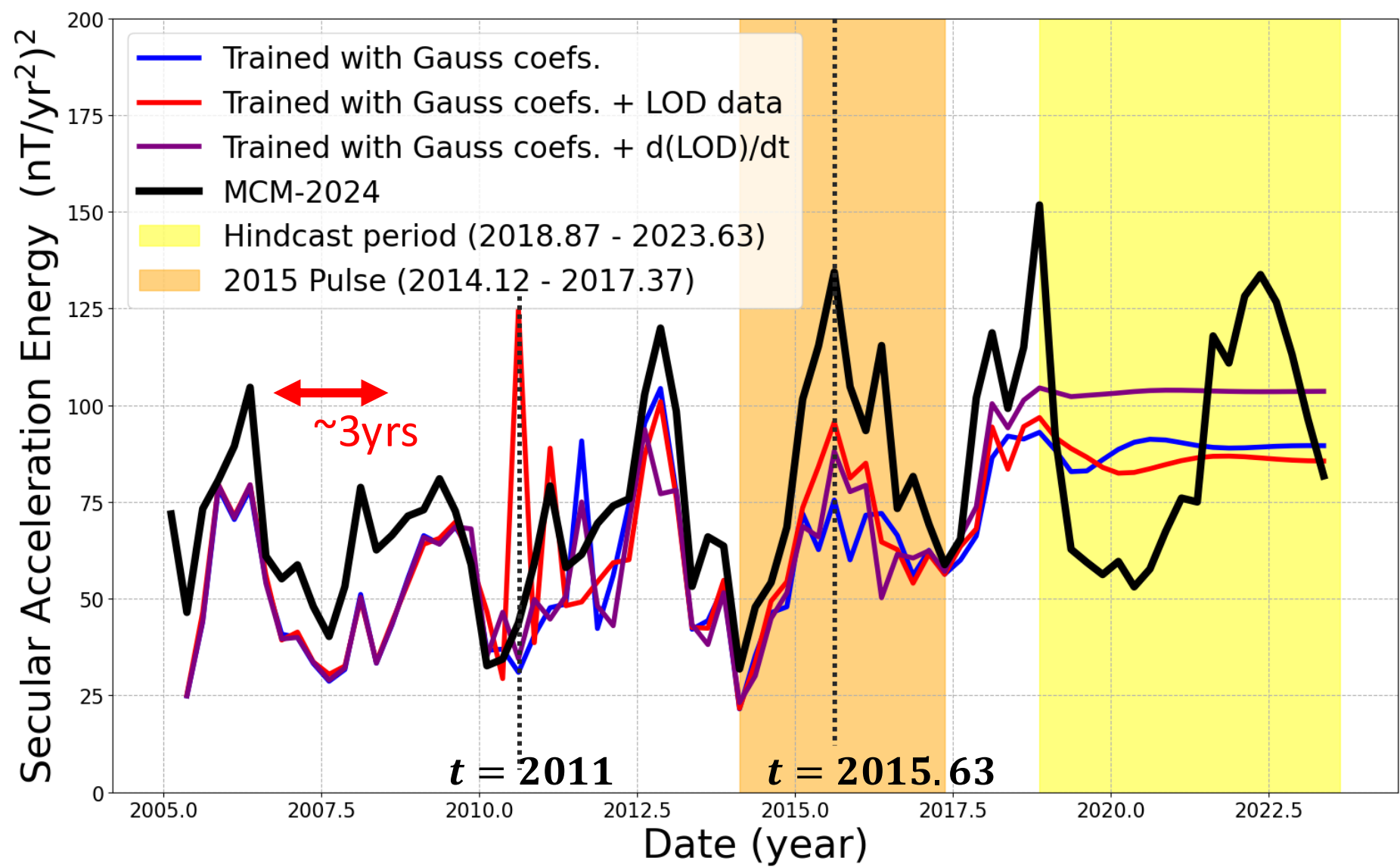
APPENDIX | SA Energy Periodicity & Azm. wave number



$$E^{m(SA)} = \sum_{\ell=m}^{13} (\ell + 1) [(g_{\ell}^m)^2 + (h_{\ell}^m)^2]$$
$$E_{\text{tot}}^{(SA)} = \sum_{m=0}^{13} \sum_{\ell=m}^{13} (\ell + 1) [(g_{\ell}^m)^2 + (h_{\ell}^m)^2]$$



APPENDIX | LDT-RNN avoided spike at t=2011



- **EKF-RNN trained only with SA snapshots** failed to learn **SA pulse in 2015** and SA patch on the equatorial region.
- **EKF-RNN with LOD-augmented training (LOD-RNN)** improved the **training misfit for the SA pulse in 2015**.
- **EKF-RNN with dLODdt-augmented training (LDT-RNN)** removed the **unnatural spike at 2011**.

